

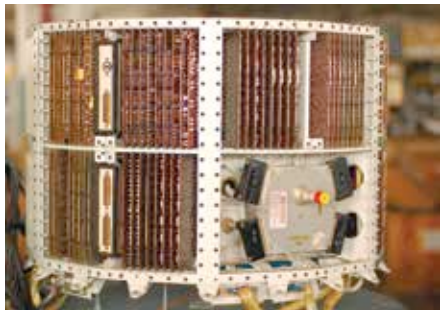
out and ask your nearest computer walla for an embedded OS on your PC/laptop. Do you really need one? Decide if you will really use it – else you'll be paying extra for something which has no function other than making your not-so-tech-savvy friends go “Wow!”.

So isn't any device from time immemorial an embedded system? Well, no. The first use of an embedded system was in 1961, and it was only in the 1980s when it was able to break into the market commercially. Before this, intelligent systems weren't possible as embedded systems weren't feasible, because of financial and logistical constraints. Read on to take a fast-track trip through the history of Embedded Systems and understand why they are such an important milestone in technological advances.

History of embedded systems

The minuteman ICBM

The Minuteman intercontinental ballistic missile was the first to use an embedded computing system, and was built in 1961 for the US Air Force. This embedded system was the Autonetics D-17 guidance computer. Embedded microprocessors, in addition to the advances in solid-fuel propellants enabled the Air Force to develop its first solid-fuel ICBM. The embedded system controlled avionics (a cool way of saying aviation and electronics in one word) and inertial guidance for the missile. The Minuteman was a 3 stage solid-propellant rocket-powered ICBM with a range of around 5,500 nautical miles. The system helped in creating a mass producible, simple and efficient ICBM capable of destroying enemy targets with consistent reliability, which is how wars are won.



The Autonetics D-17

Apollo Guidance Computer

The Apollo guidance computer was a digital computer introduced in August 1966 for the Apollo Program that was installed onboard each Apollo Command Module and Lunar Module. It was the first integrated